Week 10-11 Office Hour Notes

10.0 Errata

10.1 Structures of a Language and Examples

Recall that a language $\mathfrak{L} := (\mathcal{F}, \mathcal{R})$ consists of a set $\mathcal{F}$ of function/constant symbols and a set $\mathcal{R}$ of predicate symbols

Definition 10.1.1. A **structure** $\mathfrak{A} := \left(|\mathfrak{A}|, \left\{f^{\mathfrak{A}}\right\}_{f\in\mathcal{F}}, \left\{R^{\mathfrak{A}}\right\}_{R\in\mathcal{R}}\right)$ of $\mathfrak{L}$ (or $\mathfrak{L}$**-structure**) consists of a (usually nonempty) set $|\mathfrak{A}|$ and for every n-ary function (or constant) symbol $f \in \mathcal{F}$, an n-ary function $f^{\mathcal{A}} : |\mathfrak{A}|^n \to |\mathfrak{A}|$, and for every n-ary predicate symbol $R \in \mathcal{R}$, an n-ary relation $R^{\mathfrak{A}} \subset |\mathfrak{A}|^n$.

Remark 10.1.1. When we discuss the set of n-ary functions corresponding to $\mathcal{F}$, in these notes we often write $\mathcal{F}^{\mathfrak{A}}$, and when discussing the set of n-ary relations corresponding to $\mathcal{R}$, in these notes we often write $\mathcal{R}^{\mathfrak{A}}$.

Examples 10.1.2.

(i) Let $\mathfrak{L}_0$ be the empty language, only containing equality. The set $|\mathfrak{A}| := \{a, b\}$ for distinct $a \neq b$ would be an example of a structure in that language (note that no functions or relations need to be defined since there are no function or predicate symbols)

(ii) Let $\mathfrak{L}_{\text{set}}$ be the language of sets, consisting of $\in$ symbol. Define A_n recursively as follows

$A_0 := \varnothing,\ A_{n+1} := A_n \cup \{A_n\},$

so

$$\begin{aligned}
A_1 &= \{\varnothing\} = \{A_0\}, \\
A_2 &= \{\varnothing, \{\varnothing\}\} = \{A_0, A_1\}, \\
A_3 &= \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\} = \{A_0, A_1, A_2\} \\
&\vdots \\
A_n &= \{A_0,\ \ldots, A_{n-1}\}
\end{aligned}$$

Define

$A_\omega := \{A_n : n \in \mathbb{N}_0\} = \{A_0, A_1, A_2, \dots, A_n, \dots\}$

We find $\mathfrak{A} := \left(|\mathfrak{A}|, \in^{\mathfrak{A}}\right)$ defined by $|\mathfrak{A}| := A_\omega$,

$\in^{\mathfrak{A}} := \{(A_j, A_k) : A_j \in A_k\}$

VERY IMPORTANT NOTE: When we define structures of a given, we do so within a given "metalanguage". This metalanguage typically involves working within "Zermelo-Frankel Choice" ZFC set theory, so symbols like $\in$, $<$, etc. are used outside the language and within the metalanguage when defining structures in the language we are talking about.

So note that the structures that we define use the metalanguage to define it, so the structures themselves don't define the metalanguage. Talking about how sets and other mathematical constructs are defined and derived rigorously in a metalanguage is beyond the scope of this course and we do that more in a set theory/numerical analysis course.

(iii) In the language of number theory $(S, +, \cdot, \exp, <)$, we can define the structure $\mathfrak{A} := \left(\mathbb{N}_0, 0^{\mathfrak{A}}, S^{\mathfrak{A}}, +^{\mathfrak{A}}, \cdot^{\mathfrak{A}}, \exp^{\mathfrak{A}}, <^{\mathfrak{A}}\right)$ by

$|\mathfrak{A}| := \mathbb{N}_0 = \{0, 1, 2, \dots\}$,
$0^{\mathfrak{A}} := 0$
$S^{\mathfrak{A}}(n) := n + 1$,
$+^{\mathfrak{A}}(n, m) := n + m$,
$\cdot^{\mathfrak{A}}(n, m) := n \cdot m$,
$\exp^{\mathfrak{A}}(n, m) := n^m$,
$<^{\mathfrak{A}}(n, m) \iff n < m$,

Note again as we remarked previously that we are using what we know of the set of natural numbers, addition, multiplication, exponentiation, and inequalities in the metalanguage to define the structure $\mathfrak{A}$ in this example.

10.2 Variable Assignment Functions and Truth With Respect to a Structure and Variable Assignment

Given a language $\mathfrak{L} := (\mathcal{F}, \mathcal{R})$, let $V := \{v_k\}_{k \in \mathbb{N}}$ be the set of variables, and $\text{Term}_{\mathfrak{L}}$ be the set of terms in the language.

Definition 10.2.1. Given a structure $\mathfrak{A}$ in language $\mathcal{L}$, a **variable assignment function** is a function $s : V \longrightarrow |\mathfrak{A}|$.

By recursion on the inductive set $\text{Term}_{\mathcal{L}}$ on the function (and constant) symbols in $\mathcal{F}$ (see **Definition 7.3.1** of the Week 7 Office Hour Notes for how the set $\text{Term}_{\mathcal{L}}$ is defined) along with the variables V (refer to **Theorem 4.1.4** and **Theorem 4.2.3** of the Week 4 Office Hour Notes), we can extend the function s into a **term assignment function** $\overline{s} : \text{Term}_{\mathcal{L}} \longrightarrow |\mathfrak{A}|$ as follows:

(i) (base case) On variables v_k, we set $\overline{s}(v_k) := s(v_k)$, and on constant symbols $c \in \mathcal{F}$, we set $\overline{s}(c) := c^{\mathfrak{A}} \in |\mathfrak{A}|$.

(ii) (inductive case) Given an n-ary function symbol $f \in \mathcal{F}$, for $n \geq 1$, and $t_1, \ldots, t_n \in \text{Term}_{\mathcal{L}}$, we define

$$\overline{s}(ft_1, \ldots, t_n) = f^{\mathfrak{A}}(\overline{s}(t_1), \ldots, \overline{s}(t_n)).$$

Example 10.2.2. Let's say $\mathcal{L} := (S, +, \cdot, \exp, <)$ is the language of number theory and $\mathfrak{A} := \left(\mathbb{N}_0, 0^{\mathfrak{A}}, S^{\mathfrak{A}}, +^{\mathfrak{A}}, \cdot^{\mathfrak{A}}, \exp^{\mathfrak{A}}, <^{\mathfrak{A}}\right)$ is as defined in **Example 10.1.2** *(iii)*. Define $s : V \longrightarrow |\mathfrak{A}|$ by $s(v_n) := n$. We find that

$$\begin{aligned}
\overline{s}(+S0SSv_2) &= +^{\mathfrak{A}}(\overline{s}(S0), \overline{s}(SSv_2)) \\
&= +^{\mathfrak{A}}\left(S^{\mathfrak{A}}(\overline{s}(0)), S^{\mathfrak{A}}(\overline{s}(Sv_2))\right) \\
&= +^{\mathfrak{A}}\left(S^{\mathfrak{A}}\left(0^{\mathfrak{A}}\right), S^{\mathfrak{A}}\left(S^{\mathfrak{A}}(s(v_2))\right)\right) \\
&= +^{\mathfrak{A}}\left(S^{\mathfrak{A}}\left(0^{\mathfrak{A}}\right), S^{\mathfrak{A}}\left(S^{\mathfrak{A}}(2)\right)\right) \\
&= S^{\mathfrak{A}}\left(0^{\mathfrak{A}}\right) + S^{\mathfrak{A}}\left(S^{\mathfrak{A}}(2)\right) \\
&= \left(0^{\mathfrak{A}} + 1\right) + \left(S^{\mathfrak{A}}(2) + 1\right) \\
&= (0 + 1) + ((2 + 1) + 1) \\
&= 5.
\end{aligned}$$

Definition 10.2.3. Given a language $\mathcal{L}$, a structure $\mathfrak{A} := \left(|\mathfrak{A}|, \left\{f^{\mathfrak{A}}\right\}_{f \in \mathcal{F}}, \left(R^{\mathfrak{A}}\right)_{R \in \mathcal{R}}\right)$, and a variable assignment function $s : V \longrightarrow |\mathfrak{A}|$, and a wff ϕ, we say that ϕ **is true in a structure** $\mathfrak{A}$ **with respect to a variable assignment function** s, and write $\vDash_{\mathfrak{A}} \phi[s]$ (or in other texts, we write $\mathfrak{A} \vDash \phi[s]$) if the following recursive scenarios (on wff's in the first order setting, refer to **Section 8.2** of the Week 8 Office Hour Notes for the inductive properties on such wff's) are satisfied:

(i) (base case) If ϕ is an atomic formula of the form $= t_1 t_2$ (if the language includes equality) or $Rt_1, \ldots, t_n$ for n-ary predicate symbol $R \in \mathfrak{R}$, then

$$\vDash_{\mathfrak{A}} = t_1 t_2[s] \iff \overline{s}(t_1) = \overline{s}(t_2) \text{ (in the structure set } |\mathfrak{A}|\text{)}$$
$$\vDash_{\mathfrak{A}} = Rt_1, \ldots, t_n[s] \iff (\overline{s}(t_1), \ldots, \overline{s}(t_n)) \in R^{\mathfrak{A}}$$

(ii) (inductive case) If ϕ is a formula of the form $\neg\phi_1, \phi_1 \to \phi_2,$ or $\forall x \phi_1$, then

$\vDash_{\mathfrak{A}} (\neg\phi_1)[s]$ if $\vDash_{\mathfrak{A}} \phi_1[s]$ doesn't hold, i.e. $\nvDash_{\mathfrak{A}} \phi_1[s]$
$\vDash_{\mathfrak{A}} \left(\phi_1 \to \phi_2\right)[s]$ if either $\nvDash_{\mathfrak{A}} \phi_1[s]$ or $\vDash_{\mathfrak{A}} \phi_1[s]$ and $\vDash_{\mathfrak{A}} \phi_2[s]$.
$\vDash_{\mathfrak{A}} (\forall x \phi_1)[s]$ if, for every $a \in |\mathfrak{A}|$, we have $\vDash_{\mathfrak{A}} \phi_1[s(x|a)]$, where

$$s(x|a)(y) = \begin{cases} a & \text{if } x = y, \\ s(y) & \text{otherwise.} \end{cases}$$

%MORE RIGOROUSLY SHOW HOW THIS DEFINITION BY RECURSION WORKS

Example 10.2.4.

(i) Let $\mathfrak{L} := \mathfrak{L}_{\text{Set}}$ and $\mathfrak{A} := \left(A_\omega, \in^{\mathfrak{A}}\right)$ be as defined in **Example 10.1.2**. Let $s(v_n) := A_n$. Then

$\vDash_{\mathcal{A}} \neg \in v_2 v_1[s]$ holds since

$$\begin{aligned} \vDash_{\mathcal{A}} \neg \in v_2 v_1[s] &\iff \nvDash_{\mathfrak{A}} \in v_2 v_1[s] \\ &\iff \overline{s}(v_2) \notin^{\mathfrak{A}} \overline{s}(v_1) \\ &\iff \overline{s}(v_2) = A_2 = \{\varnothing, \{\varnothing\}\} \notin \overline{s}(v_1) = A_1 = \{\varnothing\} \end{aligned}$$

(ii) Let $\mathfrak{L} := (S, +, \cdot, \exp, <)$ is the language of number theory and $\mathfrak{A} := \left(\mathbb{N}_0, 0^{\mathfrak{A}}, S^{\mathfrak{A}}, +^{\mathfrak{A}}, \cdot^{\mathfrak{A}}, \exp^{\mathfrak{A}}, <^{\mathfrak{A}}\right)$ is as defined in **Example 10.1.2** *(iii)*. Define $s : V \to |\mathfrak{A}|$ by $s(v_n) := n$ (as in **Example 10.2.2**). We find that

$$\nvDash_{\mathfrak{A}} = v_4 + S0SSv_2,$$

since

$$\nvDash_{\mathfrak{A}} = v_4 + S0SSv_2 \iff \overline{s}(v_4) \neq \overline{s}(+S0SSv_2),$$

and we found in **Example 10.2.2** that $\overline{s}(+S0SSv_2) = 5$, which is not equal to $\overline{s}(v_4) = 4$.

10.3 Truth of Sentences and Models of Formulas

Let $\mathfrak{L} := (\mathcal{F}, \mathcal{R})$ be an arbitrary language, let $\mathfrak{A} := \left(|\mathfrak{A}|, \mathcal{F}^{\mathfrak{A}}, \mathcal{R}^{\mathfrak{A}}\right)$, where $\mathcal{F}^{\mathfrak{A}} := \left\{f^{\mathfrak{A}}\right\}_{f \in \mathcal{F}}, \mathcal{R}^{\mathfrak{A}} := \left\{R^{\mathfrak{A}}\right\}_{R \in \mathcal{R}}$ be a structure in some , let $\phi(x_1, \dots, x_n)$ be a wff with $x_1, \dots, x_n$ as all the free variables.

Theorem 10.3.1. *(Theorem 22A in page 86 of Enderton)* Assume that s_1 and s_2 are are variable assignment functions $V \to |\mathfrak{A}|$, which agree on all variables (if any) that occur free in a given wff φ, i.e., for all *free variables* $x \in V$, we have $s_1(x) = s_2(x)$. Then

$$\vDash_{\mathfrak{A}} \varphi[s_1] \iff \vDash_{\mathfrak{A}} \varphi[s_2].$$

Outline of proof. First we prove the following claim.

Claim. If $t(x_1, \dots, x_n)$ is a term in φ with the only variables $x_1, \dots, x_n$ utilized being free in φ, then

$$\overline{s_1}(t(x_1, \dots, x_n)) = \overline{s_2}(t(x_1, \dots, x_n)). \quad (10.1)$$

Proof. By induction on Term$_{\mathfrak{L}}$. The base case where $t(x_1, \dots, x_n)$ is some x_k, for $1 \leq k \leq n$, we find by the hypothesis that s_1, s_2 agree on x_k, since x_k is free in φ, we have

$$\overline{s_1}(t(x_1, \dots, x_n)) = s_1(x_k) = s_2(x_k) = \overline{s_2}(t(x_1, \dots, x_n)).$$

In the inductive step, if $t(x_1, \dots, x_n)$ is of the form $ft_1(x_1, \dots, x_n), \dots, t_m(x_1, \dots, x_n)$, for some m-ary function symbol $f \in \mathcal{F}$, and the inductive hypothesis holding that (10.1) holds for the terms $t_1(x_1, \dots, x_n), \dots, t_m(x_1, \dots, x_n)$, we find by inductive hypothesis and **Definition 10.2.1** *part (ii)* that

$$\begin{aligned}\overline{s_1}(t(x_1, \dots, x_n)) &= \overline{s_1}(ft_1(x_1, \dots, x_n), \dots, t_m(x_1, \dots, x_n)) \\ &= f^{\mathfrak{A}}\left(\overline{s_1}(t_1(x_1, \dots, x_n)), \dots, \overline{s_1}(t_m(x_1, \dots, x_n))\right)\end{aligned}$$

$$
\begin{aligned}
&= f^{\mathfrak{A}}\left(\overline{s_2}(t_1(x_1, \dots, x_n)), \dots, \overline{s_2}(t_m(x_1, \dots, x_n))\right) \\
&= \overline{s_2}(ft_1(x_1, \dots, x_n), \dots, t_m(x_1, \dots, x_n)) \\
&= \overline{s_2}(t(x_1, \dots, x_n)),
\end{aligned}
$$

and the proof of the claim is complete. □

Base case. Suppose φ is $= t_1 t_2$ or $Rt_1, \dots, t_n$, It suffice to prove $\Longrightarrow$, since the converse is a similar hypothesis. Supposing $\vDash_{\mathfrak{A}} \varphi[s_1]$, we note by the above claim that since the terms $t_1, \dots, t_n$ utilize variables that are free in φ (since φ is atomic and utilizes no quantifiers), we find that

$$\overline{s_1}(t_1) = \overline{s_2}(t_1), \dots, \overline{s_1}(t_n) = \overline{s_2}(t_n),$$

and it immeidately follows that

$$
\begin{aligned}
\vDash_{\mathfrak{A}} = t_1 t_2[s_1] &\Longrightarrow \overline{s_2}(t_1) = \overline{s_1}(t_1) = \overline{s_1}(t_2) = \overline{s_2}(t_2) \\
&\Longrightarrow \vDash_{\mathfrak{A}} = t_1 t_2[s_2],
\end{aligned}
$$

$$
\begin{aligned}
\vDash_{\mathfrak{A}} Rt_1 \dots t_n[s_1] &\Longrightarrow \left(\overline{s_1}(t_1), \dots, \overline{s_1}(t_n)\right) = \left(\overline{s_2}(t_1), \dots, \overline{s_2}(t_n)\right) \in R^{\mathfrak{A}} \\
&\Longrightarrow \vDash_{\mathfrak{A}} Rt_1 \dots t_n[s_2].
\end{aligned}
$$

Inductive step. Suppose α, β are wff's that satisfy the inductive hypothesis that

$$
\begin{aligned}
&\vDash_{\mathfrak{A}} \alpha[s_1] \Longleftrightarrow \vDash_{\mathfrak{A}} \alpha[s_2], \quad (10.2) \\
&\vDash_{\mathfrak{A}} \beta[s_1] \Longleftrightarrow \vDash_{\mathfrak{A}} \beta[s_2].
\end{aligned}
$$

Using (10.2) and **Definition 10.2.3** *part (ii)*, we can complete the proof in the following cases:

(i) In the case where φ is $\neg\alpha$, we find that

$$
\begin{aligned}
\vDash_{\mathfrak{A}} \varphi[s_1] &\Longleftrightarrow \nvDash_{\mathfrak{A}} \alpha[s_1] \\
&\Longleftrightarrow \nvDash_{\mathfrak{A}} \alpha[s_2] \\
&\Longleftrightarrow \vDash_{\mathfrak{A}} \varphi[s_2].
\end{aligned}
$$

(ii) In the case where φ is $\alpha \to \beta$, we find that

$$\vDash_{\mathfrak{A}} \varphi[s_1] \Longleftrightarrow \nvDash_{\mathfrak{A}} \alpha[s_1] \text{ or } \vDash_{\mathfrak{A}} \alpha[s_1] \text{ and } \vDash_{\mathfrak{A}} \beta[s_1]$$

$\iff \not\models_{\mathfrak{A}} \alpha[s_2]$ or $\models_{\mathfrak{A}} \alpha[s_2]$ and $\models_{\mathfrak{A}} \beta[s_2]$
$\iff \models_{\mathfrak{A}} (\alpha \to \beta)[s_2]$.

(iii) In the case where φ is $\forall x \alpha$, we find by hypothesis that for every $a \in |\mathfrak{A}|$, we find $s_1(x|a), s_2(x|a)$, agree on all free variables of α (including x), and it follows that

$$\begin{aligned} \models_{\mathfrak{A}} \varphi[s_1] &\iff \text{for every } a \in |\mathfrak{A}|, \text{ we have } \models_{\mathfrak{A}} \alpha[s_1(x|a)] \\ &\iff \text{for every } a \in |\mathfrak{A}|, \text{ we have } \models_{\mathfrak{A}} \alpha[s_2(x|a)] \\ &\iff \models_{\mathfrak{A}} \varphi[s_2]. \qquad \square \end{aligned}$$

%CHECK THIS PROOF

Corollary 10.3.2. *(Corollary 22B of Enderton page 87)* For a sentence σ, either

(a) $\models_{\mathfrak{A}} \sigma[s]$, for every variable assignment function $s : V \to |\mathfrak{A}|$; or
(b) $\not\models_{\mathfrak{A}} \sigma[s]$, for every variable assignment function $s : V \to |\mathfrak{A}|$.

Proof. We have two cases

(i) Suppose that $\not\models_{\mathfrak{A}} \sigma[s]$, for every variable assignment function s. Then condition *(b)* satisfied and we're done

(ii) Suppose that $\models_{\mathfrak{A}} \sigma[s]$, for some variable assignment function s. Then for any other variable assignment function s', since σ is a sentence with no free variables, s and s' agree on all free variables (vacuously), hence by **Theorem 10.3.1**, we have

$$\models_{\mathfrak{A}} \sigma[s] \Longrightarrow \models_{\mathfrak{A}} \sigma[s'],$$

and our conclusion is reached. $\square$

The takeaway from this result is we can talk about truth of sentences in particular purely based on the structure and regardless of variable assignment functions. Moving forward, we can state that σ (instead of just $\models_{\mathfrak{A}} \sigma[s]$) is a true sentence with respect to $\mathfrak{A}$ and write

$\models_{\mathfrak{A}} \sigma$ to mean $\models_{\mathfrak{A}} \sigma[s]$, for every variable assignment function s.

Variable assignment are only necessary in case where we deal with formulas with free variables, where truth ends up depending on what the values of the free variables end up being, which gives rise to the following definition:

Definition 10.3.3. Given a sentence σ, we state that $\mathfrak{A}$ **models** σ **if it holds that** $\vDash_{\mathfrak{A}}\sigma$.
%CONSIDER REORGANIZING THIS DEFIITION IN FUTURE DRAFTS

11.0 Errata

11.1 Logical Implication and Definability of a Structure

Let $\mathfrak{L} := (\mathcal{F}, \mathcal{R})$ be an arbitrary language, let $\mathfrak{A} := \left(|\mathfrak{A}|, \mathcal{F}^{\mathfrak{A}}, \mathcal{R}^{\mathfrak{A}}\right)$, where $\mathcal{F}^{\mathfrak{A}} := \left\{f^{\mathfrak{A}}\right\}_{f\in\mathcal{F}}, \mathcal{R}^{\mathfrak{A}} := \left\{R^{\mathfrak{A}}\right\}_{R\in\mathcal{R}}$ be a structure in some , let $\phi(x_1, \dots, x_n)$ be a wff with $x_1, \dots, x_n$ as all the free variables.

Definition 11.1.1. Let Γ be a set of wff's and ϕ be a wff, We state that Γ **logically implies** ϕ write $\Gamma \vDash \phi$ if for every structure $\mathfrak{A}$ and variable assignment function $s : V \to |\mathfrak{A}|$ such that:

$\vDash_{\mathfrak{A}}\gamma[s]$, for every $\gamma \in \Gamma$,

we have $\vDash_{\mathfrak{A}}\phi[s]$.

In the case where $\Gamma = \{\psi\}$, for some wff ψ, we write $\psi \vDash \phi$ in place of $\{\psi\} \vDash \phi$, and say that ψ **logically implies** ϕ, and if $\psi \vDash \phi$ and $\phi \vDash \psi$, we write $\psi \vDash\Dashv \phi$ and state that ψ **is logically equivalent to** ϕ.

Finally, in the case where $\Gamma = \varnothing$, we write $\vDash \phi$ in place of $\varnothing \vDash \phi$, and say that ϕ **is valid**.

Moving forward, let Γ be a set of well-formed formulas.
%AND THEN MAKE PROPOSITION ON THAT
%THEN EXPLAIN WHY THERE'S DIFFERENT NOTATIONAL CONVENTIONS AND WHY MINE IS DIFFERENT (AND MOST CERTAINLY BETTER!)
%MAKE REMARK ABOUT ALTERNATIVE CONVENTIONS OF VALIDITY

%FIGURE OUT HOW TO BETTER LABEL/SPLIT THIS LEMMA
Lemma 11.1.2. Tautological implication in first order logic extends the notion of tautological implication in sentential logic, i.e., we have:

(i) $\Gamma \vDash \neg\phi \iff$ for every structure $\mathfrak{A}$ and variable assignment function s such that $\vDash_{\mathfrak{A}}\gamma[s]$, for every $\gamma \in \Gamma$, we have $\nvDash_{\mathfrak{A}}\phi[s]$.

(ii) $\Gamma \vDash \phi$ and $\Gamma \vDash \psi$ iff $\Gamma \vDash \psi \wedge \psi$.

(iii) If $\Gamma \vDash \phi$ or $\Gamma \vDash \psi$, then $\Gamma \vDash \phi \vee \psi$, however the converse fails (recall an exercise in Homework 2).

(iv) If $\Gamma \vDash \phi \implies \Gamma \vDash \psi$, then $\Gamma \vDash \phi \to \psi$, however the converse fails (for similar reasons that the converse of *part (iii)*).

(v) $\Gamma \vDash \phi \iff \Gamma \vDash \psi$ iff $\Gamma \vDash \phi \leftrightarrow \psi$.

Proof. I'll prove some of the results later, and leave others as exercises. □

Lemma 11.1.3. if s_1 and s_2 agree for the particular free variables $x_1, \dots, x_n$, we have

$\vDash_{\mathfrak{A}} \phi(x_1, \dots, x_n)[s_1] \iff \vDash_{\mathfrak{A}} \phi(x_1, \dots, x_n)[s_2]$.

Proof. This is just a restatement (in different notation) of **Theorem 10.3.1**. □

Definition 11.1.4. If for some variable assignment function s and $a_1, \dots, a_n \in |\mathfrak{A}|$ such that $s(x_1) = a_1, s(x_2) = a_2, \dots, s(x_n) = a_n$, we have $\vDash_{\mathfrak{A}} \phi(x_1, \dots, x_n)[s]$, then we state $\vDash_{\mathfrak{A}} \phi[[a_1, \dots, a_n]]$.

Moreover, for wff's $\alpha(x_1, \dots, x_n), \beta(x_1, \dots, x_n)$ with free variables $x_1, \dots, x_n$ (or more generally if β has $x_1, \dots, x_n$ as either free variables or variables that don't appear at all) and $a_1, \dots, a_n \in |\mathfrak{A}|$, we define

$\Gamma \vDash_{\mathfrak{A}} \beta(a_1, \dots, a_n)$,

if for every variable assignment function s such that $s(x_1) = a_1, \dots, s(x_n) = a_n$ and $\vDash_{\mathfrak{A}} \gamma[s]$, for every $\gamma \in \Gamma$, we have $\vDash_{\mathfrak{A}} \beta[[a_1, \dots, a_n]]$. Moreover, if $\Gamma = \{\alpha\}$, we write

$\alpha(a_1, \dots, a_n) \vDash_{\mathfrak{A}} \beta(a_1, \dots, a_n)$.

Note that this definition makes sense (using ϕ and $\phi(x_1, \dots, x_n)$ interchangeably) by **Lemma 11.1.3**, since truth of ϕ is invariant of truth assignment functions s, s' that agree on the free variables $x_1, \dots, x_n$ of ϕ.

%COMPARE THIS APPROACH TO OTHER TEXTS

This next proposition will tell us that logical implication of quantifiers behave the way that we expect.

%GIVE PRELIMINARY PROPOSITION

Proposition 11.1.5. Suppose x is free in the wff $\phi(x)$.

(i) The following are equivalent:

(a) $\Gamma \vDash \forall x \phi(x)$

(b) For every structure $\mathfrak{B}$ and variable assignment function $s : V \longrightarrow |\mathfrak{B}|$, we find that

$\vDash {}_{\mathfrak{B}}\gamma[s(x|d)]$, for all $\gamma \in \Gamma \Longrightarrow \vDash {}_{\mathfrak{B}}\phi[s(x|d)]$, for all $d \in |\mathfrak{B}|$

(c) For every structure $\mathfrak{B}$, we have $\Gamma \vDash {}_{\mathfrak{B}}\phi[[d]]$, for all $d \in |\mathfrak{B}|$.

(ii) The following are equivalent.

(a) $\Gamma \vDash \exists x \phi(x)$

(b) For every structure $\mathfrak{B}$, $d \in |\mathfrak{B}|$, and variable assignment function s, there exists $d \in |\mathfrak{B}|$ such that $\vDash {}_{\mathfrak{B}}\gamma[s(x|d)]$, for all $\gamma \in \Gamma$ and $\vDash {}_{\mathfrak{B}}\phi[s(x|d)]$.

(c) For every structure $\mathfrak{B}$, there exists some $d \in |\mathfrak{B}|$ such that $\Gamma \vDash {}_{\mathfrak{B}}\phi[[d]]$.

Proof. I'll prove some of the results later, and leave others as exercises. ◻

Definition 11.1.6.

(i) Given a set $W \subset |\mathfrak{A}|^n$, we define

$\phi(W) := \{(a_1, \dots, a_n) \in W : \vDash {}_{\mathfrak{A}}\phi[[a_1, \dots, a_n]]\}$.

(ii) We call a subset $X \subset |\mathfrak{A}|^m$ **definable in** $\mathfrak{A}$ if for some wff $\psi(y_1, \dots, y_m)$ consisting of free variables $y_1, \dots, y_m$, we have

$X = \psi\left(|\mathfrak{A}|^m\right) = \left\{(a_1, \dots, a_m) \in |\mathfrak{A}|^m : \vDash {}_{\mathfrak{A}}\psi[[a_1, \dots, a_m]]\right\}$.

11.2 Homomorphisms and Elementary Equivalence

Let $\mathfrak{L} := (\mathcal{F}, \mathcal{R})$ be an arbitrary language, let $\mathfrak{A} := \left(|\mathfrak{A}|, \mathcal{F}^{\mathfrak{A}}, \mathcal{R}^{\mathfrak{A}}\right), \mathfrak{B} := \left(|\mathfrak{B}|, \mathcal{F}^{\mathfrak{B}}, \mathcal{R}^{\mathfrak{B}}\right)$ (analogous to how notation was previously defined) with respect to a language $\mathfrak{L} := (\mathcal{F}, \mathcal{R})$.

Definition 11.2.1. We call a function $\Phi : |\mathfrak{A}| \to |\mathfrak{B}|$ a **homomorphism between the structures** $\mathfrak{A}, \mathfrak{B}$ (and write $\Phi : \mathfrak{A} \to \mathfrak{B}$) if it satisfies the following properties:

(i) for every n-ary function symbol $f \in \mathcal{F}$, and $a_1, \ldots, a_n \in |\mathfrak{A}|$ we have

$$\Phi\left(f^{\mathfrak{A}}(a_1, \ldots, a_n)\right) = f^{\mathfrak{B}}(\Phi(a_1), \ldots, \Phi(a_n)).$$

(ii) For every n-ary relation $R \in \mathcal{R}$, and $a_1, \ldots, a_n \in |\mathfrak{A}|$, we have

$$(a_1, \ldots, a_n) \in R^{\mathfrak{A}} \Longrightarrow (\Phi(a_1), \ldots, \Phi(a_n)) \in R^{\mathfrak{B}}.$$

If Φ is also one-to-one and onto (i.e. bijective), then Φ is called an **isomorphism**, and we write $\mathfrak{A} \cong \mathfrak{B}$ if there exists an isomorphism between $\mathfrak{A}$ and $\mathfrak{B}$. An isomorphism between the structure $\mathfrak{A}$, itself $\Psi : \mathfrak{A} \to \mathfrak{A}$ is called an **automorphism**.
%TALK ABOUT ENDOMORPHISMS
%TALK ABOUT WHY MARKUS AND I'S DEFINITION OF "ISOMORPHISM" IS BETTER

Note that this corresponds to definitions of homomorphisms that you might have abstract math classes such as abstract algebra and group theory, i.e., a homorphism of groups is exactly a homomorphism between the structures in $\mathfrak{L}_{\text{groups}}$, a homomorphism of graphs is exactly a homomorphism between the structures in $\mathfrak{L}_{\text{graphs}}$, etc. (as you'll see in examples in the book/powerpoint, and I may give additional examples in a later draft of these notes).

You may reallize that since formulas in logic talk about terms and relations between them, homomorphisms also preserve the statements about terms, as the following theorem states.

Theorem 11.2.2. *(The Homomorphism Theorem; as stated in page 96 the book)* Let $\Phi : \mathfrak{A} \to \mathfrak{B}$ be a homomorphism, and let $s : V \to |\mathfrak{A}|$.

(a) $\Phi \circ s : V \to |\mathfrak{B}|$ is a variable assignment function.

(b) For any term t in $\mathfrak{L}$, the homomorphism Φ preserves assignment of terms by $\overline{s}$, i.e., we have

$\Phi(\overline{s}(t)) = \overline{(\Phi \circ s)}(t)$.

(c) For any quantifier-free formula α not containing the equality symbol, then

$$\models _{\mathfrak{A}}\alpha[s] \Longleftrightarrow \models _{\mathfrak{A}}\alpha[\Phi \circ s], \ (11.1)$$

and additionally for any $a_1, \ \dots, a_n \in |\mathfrak{A}|$, and any wff $\phi(x_1, \ \dots, x_n)$ not containing equality with free variables $x_1, \ \dots, x_n$, we have

$$\models _{\mathfrak{A}}\phi[[a_1, \ \dots, a_n]] \Longleftrightarrow \models _{\mathfrak{A}}\phi[[\Phi(a_1), \ \dots, \Phi(a_n)]. \ (11.2)$$

(d) If Φ is one-to-one (i.e. injective), then (11.1) and (11.2) for any α and $\phi(x_1, \ \dots, x_n)$ contains the equality symbol but is quantifier free.

(e) If Φ is onto (i.e. surjective), then (11.1) and (11.2) for any α and $\phi(x_1, \ \dots, x_n)$ that contains is quantifier-free, but contains, but contains the equality symbol.

Outline of the poof.

(a) Immediate by applying the composition of functions to the definition of a variable assignment function (**Definition 10.2.1**).

(b) Straightforward structural induction on the term t using *property (i)* of **Definition 10.3.1**.

(c)-(e) More or less a corollary to the following reiteration of the Homomorphism Theorem: □

Theorem 11.2.3. *(A Reiteration of the Homomorphism Theorem)* Let $\Phi : \mathfrak{A} \rightarrow \mathfrak{B}$ be an *onto* homomorphism, and let $s : V \rightarrow |\mathfrak{A}|$.

(a) For any wff α, we have

$$\models _{\mathfrak{A}}\alpha[s] \Longrightarrow \models _{\mathfrak{A}}\alpha[\Phi \circ s],$$

and additionally for any $a_1, \ \dots, a_n \in |\mathfrak{A}|$, and any wff $\phi(x_1, \ \dots, x_n)$ not containing equality with free variables $x_1, \ \dots, x_n$, we have

$$\models _{\mathfrak{A}}\phi[[a_1, \ \dots, a_n]] \Longrightarrow \models _{\mathfrak{A}}\phi[[\Phi(a_1), \ \dots, \Phi(a_n)]. \ (11.2)$$

(b) If $\Phi : \mathfrak{A} \to \mathfrak{B}$ is also one-to-one, then its inverse $\Phi^{-1} : |\mathfrak{B}| \to |\mathfrak{A}|$ (recall that a function is invertible iff it is one-to-one and onto) is an isomorphism $\mathfrak{B} \to \mathfrak{A}$.

Outline of proof.

(a) Straightforward structural induction on wff's α and $\phi(x_1, \ldots, x_n)$ and application of part (b) of the previous Theorem. (will update specifics in another draft)

(b) Straightforward application of Definition 11.3.1, along with the fact that it's one-to-one. (will update specifics in another draft) □

Definition 11.2.4. We call structures $\mathfrak{A}, \mathfrak{B}$ elementarily equivalent and write $\mathfrak{A} \equiv \mathfrak{B}$ if for every sentence σ, we have

$$\vDash_{\mathfrak{A}} \sigma \iff \vDash_{\mathfrak{B}} \sigma.$$

Corollary 11.2.5. $\mathfrak{A} \cong \mathfrak{B} \Longrightarrow \mathfrak{A} \equiv \mathfrak{B}$.

Proof. Immediate from part *(d)* and *(e)* of **Theorem 11.2.2.** □

11.3 Free Structures

You see a lot a "free" [random math object here] "generated by" [insert set]. The terminology comes up in this course due to (long story short) the generalized notion of induction and recursion that we learned in Week 4 being related to such a free object construction.

Defining this in a completely rigorous way, as opposed to a handwavy way, unfortunately utilizes category theory that goes beyond the scope of this course (in a later draft I'll link the category theoretic idea, but again it's very dense mathematal material that takes a graduate exposition of algebra/category theory to truly understand).

However, I'll do the best I can and define a free structure $\mathfrak{A}_S$ generated by a set S, in a language $\mathcal{L} = (\mathcal{F})$, with no relations.

Definition 11.3.1. A **free structure** $\mathfrak{A}_S$ **generated by a set** S is defined to be the "words" utilizing elements $s \in S$ in place of where variables usually go, and n-ary function symbol $f \in \mathcal{F}$ that take for every $t_1, \ldots, t_n \in |\mathfrak{A}_S|$ and defines $ft_1 \ldots t_n \in |\mathfrak{A}_S|$. It's defined more rigorously via structural induction to be the smallest set $|\mathfrak{A}_S| := \text{Term}_{\mathcal{L}}(S)$ containing S and for every $f \in \mathcal{F}$, closed under the set-builder function $\mathcal{E}_f$, which recall is the n-ary function that takes words $t_n, \ldots, t_n$ and maps it to the word $\mathcal{E}_f(t_1, \ldots, t_n) = ft_1 \ldots t_n$.

So note that $\text{Term}_{\mathcal{L}}(S)$ has the following structural inductive property:

(i) (Base case): All $s \in S$ or the constant (0-ary function) symbols $c \in \mathcal{F}_0$.

(ii) (Inductive step): For every $t_1, \ldots, t_n \in S$ and n-ary function symbol $f \in \mathcal{F}$, we have closure under set builder function $\mathcal{E}_f$ and $ft_1 \ldots t_n$ becomes included.

Note that a free set generated by a structure S, ends up having the following universal mapping property: For all functions $f : S \to |\mathfrak{B}|$ for some structure $\mathfrak{B}$, there exists a unique structure homomorphism $\phi_f : \mathfrak{A}_S \to \mathfrak{B}$ such that $i \circ \phi_f = f$.

$$\begin{tikzcd} S \arrow[r, "i"] \arrow[d, "\forall f"] & {|\mathfrak{A}_S|} \arrow[dl, "\exists! \phi_f"] \\ {|\mathfrak{B}|} & \end{tikzcd}$$

The existence and universal mapping property of this "free structure" follows precisely from **Definition by Recursion** given in week 4; in fact it's precisely "Definition by Recursion" when constructing the inductive set.

NOTE: In order for this "free structure" construction to work, we need the functions (in this case the set-builder functions) $\mathcal{E}_f$ for all $f \in \mathcal{F}$ to be injective (which thankfully they are), otherwise this explicit construction that satisfies the given universal mapping property, let alone existence could fail.

11.4 Questions